

Nested Equity Index Correlations Overstate True Co-Movement: Evidence from Vietnam

Abstract: Asset allocators in emerging Southeast Asian markets often treat large-cap and broad-market indices as separate diversification instruments, even in nested architectures where large-cap stocks dominate the broader market. Risk models built on index-level correlations therefore conflate mechanical constituent overlap with economic co-movement. Using Detrended Cross-Correlation Analysis (DCCA) and its multifractal extension on 30-minute, 1-hour, 4-hour and daily data from the Ho Chi Minh City Stock Exchange (2014–2025), this study shows that shared capitalization inflates the near-perfect correlations among nested indices (0.975–0.980). Purging constituent overlap with a capitalization-weighted mid-cap proxy lowers the correlation with large caps by 0.086–0.096, with block-bootstrap intervals that exclude zero at every frequency, although the purged correlation remains high (about 0.88). Correlations involving the broad market rise with the horizon at the 30-minute and 1-hour frequencies, whereas the purged mid-cap correlation shows no significant horizon dependence. Raw cross-tier correlation rises from 0.73 to 0.93 in high-volatility regimes, but after Forbes–Rigobon volatility conditioning there is no evidence of a structural increase, consistent with interdependence rather than contagion. A static correlation overstates in-sample mid-cap portfolio variance by 2.2–9.4% in calm regimes and understates it by about 2% in turbulent regimes. These findings support independent mid-cap benchmarks and volatility-regime-sensitive risk budgeting in Vietnam, and they motivate tests in other markets of the Association of Southeast Asian Nations (ASEAN) with nested index architectures.

Keywords: constituent overlap, DCCA, Forbes–Rigobon adjustment, multiscale correlation, portfolio risk, Vietnam.

JEL Classification: C58, G11, G12.

1 Introduction

Portfolio managers in emerging equity markets often build multi-capitalization portfolios as if large-cap and mid-cap indices were separate diversification instruments. Many risk models, asset allocation algorithms and stress tests describe cross-asset dependence with one static Pearson correlation estimated from daily closing prices (Markowitz 1952). This practice rests on two assumptions that are rarely tested: that co-movement does not depend on the holding horizon, from intraday intervals to several months, and that dependence is the same in calm and volatile regimes.

In nested equity index systems, a static correlation also mixes two sources of dependence. Across the equity markets of the Association of Southeast Asian Nations (ASEAN), index providers use nested, capitalization-tiered architectures. For example, the FTSE Bursa Malaysia KLCI (top 30) is nested within the FTSE Bursa Malaysia Top 100 Index and FBM EMAS Index, the SET50 is nested within the SET100 in Thailand, and the IDX30 is nested within the LQ45 in Indonesia. Bursa Malaysia permits Regulated Short Selling (RSS) and Intraday Short Selling (IDSS) on

eligible constituents, whereas Vietnam prohibits cash short selling (Section 1.1). On the Ho Chi Minh City Stock Exchange (HOSE), the flagship large-cap index (VN30) is contained in the top-100 benchmark (VN100), which is in turn nested within the broad market index (VNINDEX), satisfying the strict subset relation:

$$\text{VN30} \subset \text{VN100} \subset \text{VNINDEX}. \quad (1)$$

Because the 30 largest firms represent over two-thirds of the free-float market capitalization of VN100, the return co-movement among these indices is high by construction, and reading their correlation as economic linkage confuses index design with market co-movement. An investor who splits a position between VN30 and VN100 funds gains little diversification.

To separate multiscale dynamics from nonstationary noise, econophysics and financial econometrics use Detrended Cross-Correlation Analysis (DCCA) (Podobnik and Stanley 2008) and its multifractal generalization (MF-DCCA) (Zhou 2008). Building on Detrended Fluctuation Analysis (Peng et al. 1994) and its multifractal generalization (Kantelhardt et al. 2002), DCCA evaluates cross-correlations across non-overlapping temporal segments after filtering out local polynomial trends. Zebende (2011) normalized the DCCA covariance to establish a bounded correlation metric, while Podobnik et al. (2011) proposed statistical tests for power-law cross-correlations. Subsequent advances have integrated moving-average formalisms (Jiang and Zhou 2011) and transfer entropy (Zhou et al. 2025). Applied studies show that multiscale cross-correlations vary systematically across holding periods in developed equity markets, commodity baskets and cross-border assets (Al Rababa'a et al. 2021; Aslam et al. 2023; Chen et al. 2024). However, this literature mostly analyzes asset pairs with no common constituents (such as stock-bond pairs or cross-country equity benchmarks). Multiscale cross-correlations within nested index architectures, where mechanical overlap and economic transmission coexist, have received little attention.

The overlap problem arises from index design. When index providers build size-based benchmarks with top-down capitalization cut-offs, large-cap stocks dominate the broad-market index. Michis (2022) applied wavelet partial correlations to remove redundant factor exposures in non-overlapping asset clusters, and Rodriguez and Alvarez-Ramirez (2021) documented time- and scale-varying volume–return cross-correlations in US equity markets, but neither study addresses overlapping constituents. A risk model that does not purge shared capitalization mistakes index construction for economic diversification.

Market regime shifts and microstructure frictions complicate this dependence. Cross-asset correlations are asymmetric and increase during market downturns, when diversification matters most (Ang and Chen 2002; Longin and Solnik 2001). However, Forbes and Rigobon (2002) showed that unadjusted sample correlations rise with market volatility, so that an observed correlation surge may reflect volatility expansion rather than a structural change in cross-market linkages. At intraday intervals, this dynamic intersects with the Epps (1979) effect, where non-synchronous transactions and quote delays depress short-horizon correlations. In parallel, gradual information diffusion (Hong and Stein 1999) implies that news is incorporated into prices with a delay, so co-movement between securities that are repriced at different speeds can rise with the horizon. Recent evidence from Vietnam points to delayed price adjustment under retail-heavy participation (Tran and Tran 2025) and to nonlinear dependence on regional markets (Le et al. 2025).

Regional studies show that financial integration across ASEAN equity markets is time-varying and shaped by trade linkages, market volatility and external shocks (Karim and Ning 2013; Lean and Teng 2013). Because the HOSE prohibits cash short selling, unlike markets with regulated short selling, we treat the role of trading restrictions in multiscale price synchronization as a hypothesis for comparative work rather than as a tested result.

This study examines VN30, VN100 and VNINDEX from January 2014 to December 2025 at four trading frequencies: 30-minute (M30), 1-hour (H1), 4-hour (H4) and daily (1D). To separate economic linkage from constituent overlap, we construct a capitalization-weighted proxy (P_{cap}) that extracts the mid-cap component of VN100 using free-float market weights, and we benchmark it against three alternative statistical decompositions. In addition, we examine regime-dependent correlation shifts using the conditioning framework of Forbes and Rigobon (2002) and quantify the resulting misstatement in Markowitz portfolio variance.

We make three contributions to the empirical finance and risk management literature.

First, we quantify the overlap distortion in nested benchmark systems. Headline large-cap and broad-market indices show near-perfect cross-correlations (0.975–0.980) at all frequencies, and extracting shared capitalization through a market-weighted mid-cap proxy lowers the correlation with large caps by 0.086–0.096, a gap whose bootstrap confidence intervals exclude zero at every frequency. Correlations involving the broad market rise with the horizon at the 30-minute and 1-hour frequencies, whereas the purged mid-cap correlation is statistically flat across horizons.

Second, we provide a volatility-conditioned evaluation of crisis dynamics. Raw cross-tier correlations tighten by up to 0.20 correlation points in high-volatility regimes, which is usually read as a correlation breakdown. Applying the Forbes and Rigobon (2002) adjustment under two regime definitions with block-bootstrap inference, we find no evidence that the adjusted crisis correlation exceeds its calm-period level. The loss of mid-cap diversification during Vietnamese market stress is therefore consistent with interdependence amplified by volatility, although the test cannot exclude small structural increases.

Third, we quantify the risk-management cost of static correlations. Cash portfolios formed from parent and child indices show in-sample variance misstatements of at most 2.4%, whereas for a mid-cap factor exposure a static correlation overstates portfolio variance by 2.2–9.4% in calm regimes and understates it by about 2% in turbulent regimes, when volatility itself has multiplied. Because short selling is prohibited, the non-investable mid-cap construct serves as a shadow benchmark for risk attribution, stress testing and index-product development.

The remainder of the paper is organized as follows. Section 1.1 describes the current situation and institutional background of the HOSE. Section 2 presents the data and methods, Section 3 reports the results and robustness checks, and Section 4 discusses the findings, their implications and limitations, and concludes.

1.1 Current Situation and Institutional Background of the HOSE

The institutional architecture of the HOSE, regulated by the State Securities Commission of Vietnam (SSC), imposes regulatory constraints that shape price discovery, return distributions and cross-asset correlations.

The primary microstructure constraint is the symmetric daily price fluctuation band of $\pm 7\%$ relative to the reference price of the previous session. During sharp market corrections, heavily sold stocks can reach the lower limit, where trading dries up. This price collar truncates return tails and can induce serial correlation and synchronized limit hits across index constituents.

The second structural feature is the settlement cycle. Under the T+2 cycle in force for most of the sample, equities purchased on the HOSE could not be sold before the afternoon of the second business day. This settlement delay restricts intraday statistical arbitrage, preventing high-frequency traders from eliminating the pricing discrepancies between index constituents and synthetic baskets.

In addition, each trading day opens and closes with a call auction: a 15-minute batch auction sets the opening price (ATO, 09:00–09:15) and another sets the closing price (ATC, 14:30–14:45), with continuous order matching in between. At the 30-minute frequency, the first bar of each day contains the opening auction and the last bar contains the closing auction, so these bars display higher volatility and less synchronous clearing than midday bars.

Finally, securities legislation, including Decree 155/2020/ND-CP and SSC Circular 120/2020/TT-BTC, prohibits cash short selling: investors may sell only securities already held in their accounts. Margin regulations require an initial margin of at least 50%, capping gross leverage at about 2.0 $\times$, and Circular 68/2024/TT-BTC, which removed the pre-funding requirement for foreign institutional investors, left the short-sale prohibition unchanged. Although VN30 index futures trade on the Hanoi Stock Exchange derivatives market, no futures contract is listed on a mid-cap index. The absence of cash short selling prevents arbitrageurs from trading the spread between large caps and mid-caps, which can prolong intraday non-synchronicity (the Epps effect) and delay price synchronization between VN30 and mid-cap stocks.

2 Methods

2.1 Data and Sample Construction

The dataset comprises closing prices of three HOSE indices exported from TradingView (exchange: HOSE) for January 27, 2014, to December 12, 2025: the flagship VN30 index, comprising the 30 largest and most liquid stocks screened for free-float capitalization; the VN100 index, which combines the VN30 constituents with the next 70 largest liquid stocks (the constituents of the VNMIDCAP, or VN70, index); and the composite VNINDEX, which covers all common stocks listed on the HOSE and is weighted by full market capitalization. Because a VNMIDCAP price history is not available from our data source for the full sample and all frequencies, the mid-cap segment is recovered from VN100 and VN30 by the decomposition in Section 2.2.

The analysis covers four trading frequencies: 30-minute (M30), 1-hour (H1), 4-hour (H4) and daily (1D). We synchronize the index series by exact inner joins on timestamps and compute log returns as below; no index price is filled or interpolated:

(2)

We use two sample tiers.

The first tier is the Multi-Frequency Synchronized Overlap Window (January 3, 2017 to December 9, 2024), a common calendar window in which all four frequencies overlap (1D: N=1,983; M30: N=19,463; H1: N=9,879; H4: N=3,953). This sample forms the primary benchmark for cross-frequency comparisons.

The second tier is the Extended Historical Depth Sample, used for scaling regressions, regime tests and Monte Carlo reliability thresholds (1D: February 2014–December 2025, N = 2,963; M30: January 2017–December 2025, N = 21,942; H1 and H4: January 2014–December 2024, N = 13,523 and 5,410). The 1-hour and 4-hour series available from the data vendor end on December 9, 2024, whereas the daily and 30-minute series extend to December 12, 2025.

To limit survivorship and look-ahead bias, the analysis relies on official index levels calculated in real time by the HOSE rather than on indices back-calculated from current constituent lists. The HOSE reviews index constituents semi-annually, updating memberships and free-float adjustments on the basis of capitalization, liquidity and foreign-ownership criteria. Because the price series are the exchange-computed index levels, historical constituent changes, corporate actions and rebalancing adjustments are embedded in the return paths.

We evaluate regime dependence under two definitions based on the 20-day rolling standard deviation of VNINDEX returns (Fig. 1).

The first definition uses continuous chronological crisis episodes. To avoid arbitrary selection, an episode must satisfy three conditions: (i) a peak-to-trough VNINDEX drawdown of more than 25%; (ii) at least 45% of its trading days in the top quartile of the rolling-volatility distribution; and (iii) a documented macro-financial trigger. Three episodes qualify: the 2018 margin contraction (January–December 2018, including the run-up to the April peak; drawdown 26.2%, 47% of days in the top quartile), the 2020 COVID-19 shock (January–June 2020; 33.5%, 52%) and the 2022 corporate bond liquidity freeze (April–November 2022; 40.2%, 60%), totaling 539 crisis trading days. The calm benchmark comprises 2016–2017 and 2023–2024 (1,000 trading days). Transient spikes, such as the May 2014 South China Sea standoff (drawdown 14.8%) and the April 2025 tariff shock (18.1%), fail conditions (i) and (ii). The second definition sorts all days by rolling volatility, classifying days below the 25th percentile (10.6% annualized) as low-volatility and days above the 75th percentile (20.0%) as high-volatility.

Figure 1 maps the rolling annualized volatility of the VNINDEX alongside the three crisis episodes and the two quartile thresholds.

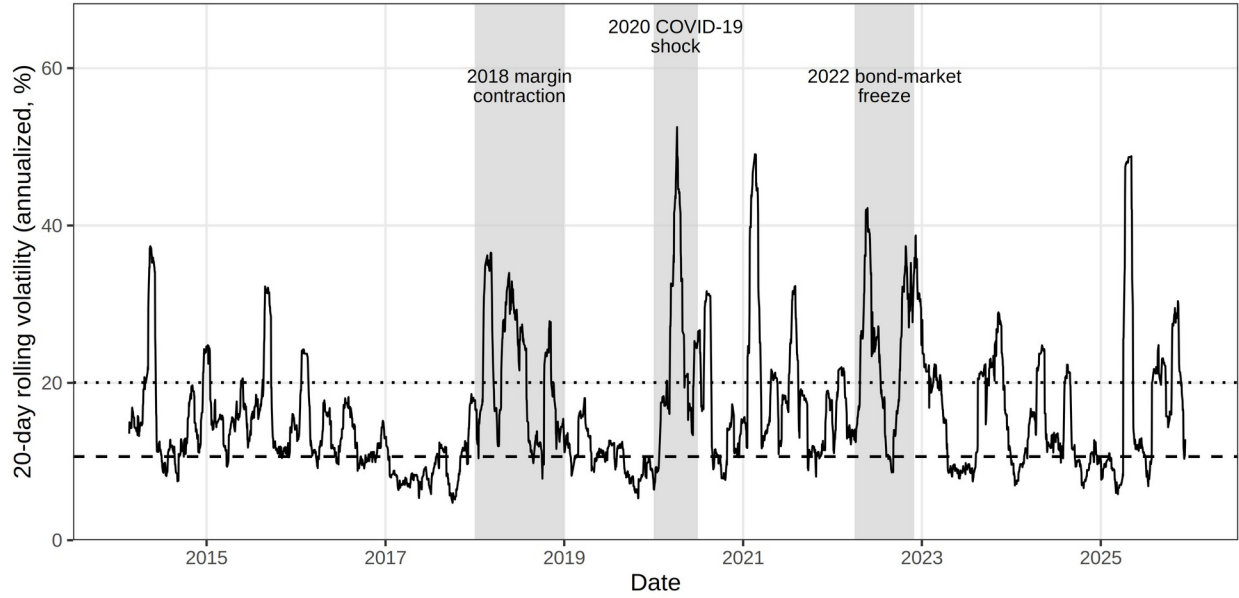


Fig. 1 Twenty-day rolling volatility of VNINDEX (annualized) and market regimes, 2014–2025

Notes: Shaded areas mark the three crisis episodes; dashed and dotted lines mark the 25th and 75th percentiles. Source: Authors' calculations based on HOSE index data (TradingView).

2.2 The DCCA Cross-Correlation Coefficient

Following Podobnik and Stanley (2008) and Zebende (2011), the DCCA cross-correlation coefficient evaluates multiscale covariance after filtering out non-stationary local trends from the data. We map the raw return series x_t and y_t into cumulative deviation profiles:

(3)

where $\bar{x}$ and $\bar{y}$ denote the respective sample means.

Both profiles are then partitioned into N non-overlapping segments of equal timescale τ . To prevent data loss near the boundary when $N\tau$ is not an exact multiple of N , the partitioning is repeated from the opposite end of the series, generating $2N$ total segments.

Within each segment, we fit local polynomial trends $\tilde{X}$ and $\tilde{Y}$ of order m using ordinary least squares (OLS; $m = 1$ representing linear DCCA) and compute the detrended covariance using piecewise boundary indexing:

(4)

To isolate aggregate co-movement at a specific holding horizon, we average the detrended covariances across all $2N_s$ segments to obtain the bivariate DCCA fluctuation function:

(5)

When $\tau = 1$, the fluctuation function collapses to the standard univariate Detrended Fluctuation Analysis (DFA) functions, and (Peng et al. 1994; Kantelhardt et al. 2002).

To bound the dependence metric within $[-1, 1]$ and make estimates at different scales comparable with ordinary correlations, the DCCA cross-correlation coefficient normalizes the bivariate covariance by the product of univariate root mean square fluctuations:

(6)

where σ_{VN30} denotes the univariate root-mean-square fluctuation function.

Because VN30 is embedded within VN100, an auxiliary regression of VN100 returns on VN30 returns produces an R^2 of 0.973–0.977 and a slope of 0.968–0.975 across the four frequencies (daily estimates shown):

(7)

To isolate the non-overlapping mid-cap return component, we evaluate four candidate proxy formulations.

First, we define a capitalization-weighted proxy (P_{cap}) based on the constituent free-float market capitalization:

(8)

where $w_{30} = 0.6826$ is the free-float market-capitalization weight of VN30 within VN100 computed from the HOSE factsheet of May 31, 2024 (VND 1,316,288 billion and VND 1,928,303 billion, respectively), implying a scaling factor of 3.151. Because the decomposition is applied to logarithmic rather than arithmetic returns, it carries a small Jensen-inequality term; at the daily frequency this term is 3.6×10^{-6} per day and, accumulated over the reliable daily horizons (up to 50 days), remains below 2.5% of the corresponding cumulative return variance, so its effect on the DCCA estimates is negligible.

For comparison, we examine a statistically weighted proxy (P_{heur}) that replaces the capitalization weight with the volatility-scaled OLS slope:

$$P_{\text{heur},t} = (R_{100,t} - w_{\text{heur}}R_{30,t}) / (1 - w_{\text{heur}}), \quad w_{\text{heur}} = \beta_{\text{OLS}}\sigma_{30}/\sigma_{100} = \text{Corr}(R_{30}, R_{100}). \quad (9)$$

Because the correlation between VN30 and VN100 returns approaches unity (0.9865–0.9885 across frequencies), the denominator collapses toward zero (0.0115–0.0135), amplifying the numerator by a factor of 74 to 87. P_{heur} therefore behaves as an amplified residual rather than an economic asset proxy: its daily standard deviation is 0.1576, compared with 0.0124 for P_{cap} . In this nested system, statistical weights are a poor substitute for capitalization weights.

Finally, we construct two baseline statistical proxies for completeness: the ratio-spread proxy (P_{ratio}), defined as the change in the log ratio of VN100 to VN30 and therefore equal to the VN100 return minus the VN30 return, and the orthogonal residual proxy (P_{res}), defined as the residual of the auxiliary regression above. The three statistical proxies are almost perfectly correlated with one another (0.980–0.997).

Because synthesizing P_{cap} requires a long position of 315% in VN100 and a short position of 215% in VN30, the proxy cannot be executed in cash portfolios under the short-sale prohibition. We therefore treat P_{cap} as an analytical shadow benchmark that isolates inter-tier economic dependence for risk budgeting.

2.3 Rebalancing and Weight Dynamics

A key question for the capitalization-weighted proxy P_{cap} is the stability of the constituent free-float market weight w_{30} over the 2014–2025 sample. The HOSE conducts semi-annual constituent reviews. During these reviews, constituent memberships are refreshed, free-float factors () are recalibrated, and individual stock weights are adjusted to enforce the index capping rules. Between these discrete semi-annual rebalancing dates, constituent market capitalizations drift with relative price movements.

When the true historical large-cap weight drifts from the static baseline benchmark, the return on the proxy extracted using fixed weights satisfies:

(10)

which introduces a time-varying leakage of large-cap returns into the proxy, proportional to the gap between the true and the baseline weight. Because only one factsheet snapshot of free-float weights is available, we do not reconstruct the historical weight path; instead, Section 3.8.2 re-estimates the key statistics for weights between 0.60 and 0.75, a range that brackets plausible drift around the baseline.

2.4 The Forbes–Rigobon Volatility Conditioning Framework

To test whether the rise in cross-correlation during turbulent periods reflects a structural change or volatility expansion, we apply the conditioning framework of Forbes and Rigobon (2002).

Let asset returns follow, with and cross-market variance. The unadjusted sample correlation is:

(11)

As market volatility rises from a calm to a crisis regime, the unadjusted correlation moves toward one because the variance of the conditioning market rises, even when the structural slope and residual variance remain constant.

To eliminate this conditioning bias, Forbes and Rigobon define the unconditional correlation as follows:

(12)

The Forbes–Rigobon identifying assumptions require that the conditioning benchmark be exogenous, with no contemporaneous feedback from the dependent series. This condition is plausible for the disjoint pair P_{cap} –VN30, in which large-cap returns serve as the market benchmark and the purged mid-cap component shares no constituents with VN30. For nested pairs (VN30–VN100, VN30–VNINDEX and VN100–VNINDEX), the assumption is violated by construction because VN30 accounts for over two-thirds of VN100 capitalization, so conditioning formulas yield unidentified parameters. Regime correlations are Pearson correlations of daily returns within each regime, and δ is the relative increase in the variance of VN30 returns, so that the correction and the correlation refer to the same measure. We restrict the formal test of a structural correlation increase to the P_{cap} –VN30 pair.

The framework has an identification limit under acute stress. Severe downturns on the HOSE can trigger retail margin calls and synchronized liquidations across large-cap and mid-cap stocks. This two-way selling pressure weakens the exogeneity condition, so we interpret the adjusted correlation as cross-tier dependence under liquidity-constrained stress, not as one-way shock transmission from large caps to mid-caps.

Inference uses a stationary block bootstrap (mean block length 20 trading days) applied separately within each regime; each of 1,999 replications recomputes the calm and crisis correlations and δ .

We report a percentile 95% interval for the difference between the adjusted crisis correlation and the calm correlation, a one-sided bootstrap p-value for the null of no increase, and the power to detect an increase of 0.05.

2.5 Log-Linear Scale Regressions and the Epps Effect

To evaluate how cross-correlation scales with the investment horizon, we regress the DCCA coefficient of each index pair at each trading frequency on the logarithm of the timescale; the slope coefficient is the multiscale scaling slope:

(13)

with the timescale evaluated on 30 logarithmically spaced points between 5 bars and one quarter of the sample length (5 to 5,485 bars at M30), and separately on the reliable range of Section 3.3. Because the DCCA coefficients at different scales are functionals of the same two series, the regression residuals are not independent across scales. Inference therefore resamples the data rather than the scales: a stationary block bootstrap of the bivariate return series (499 replications, mean block length of about 20 trading days) recomputes the whole DCCA curve and its slope in each replicate, and we report percentile 95% intervals. The same bootstrap provides the intervals for the average correlations in Table 2.

We also account for the Epps (1979) effect: at high frequencies (M30, H1), non-synchronous transactions between liquid VN30 large caps and less liquid mid-caps depress the correlation at small timescales (roughly ten bars or fewer). As the timescale increases, aggregation synchronizes price discovery and can produce a positive scaling slope; gradual information diffusion (Hong and Stein 1999) can have the same effect. Section 3.7 tests whether such slopes are present.

2.6 The MF-DCCA Extension

To capture multifractal heterogeneity across fluctuation magnitudes, we implement the MF-DCCA of Zhou (2008) using absolute local covariances to avoid complex-valued moments, an issue discussed by Oświęcimka et al. (2014), incorporating bivariate moving-average formalisms (Jiang and Zhou 2011). The q -th order fluctuation function is:

(14)

Because $q = 0$ is excluded from the grid, no logarithmic limit is required; segments with zero local covariance are discarded. The generalized cross-correlation exponent $h(q)$ is the slope of the log fluctuation function against the log timescale over 24 logarithmically spaced scales. Through the Legendre transform, the singularity strength α and the spectrum $f(\alpha)$ are:

(15)

The range of generalized exponents, $\Delta h = h(-5) - h(5)$, measures how strongly the scaling differs between small and large fluctuations. Because fat tails alone widen this range, we compare each observed range with 100 surrogates in which the paired returns are jointly shuffled, which preserves the return distributions and their contemporaneous correlation but destroys temporal structure.

3 Results

3.1 Summary Statistics

Table 1 presents the descriptive statistics for the index log returns across the four trading frequencies (2014–2025).

Table 1 Descriptive statistics of log returns at the four trading frequencies

Index	Freq.	N	Mean	Std. Dev.	Skew.	Kurt.	Jarque–Bera
VN30	Daily	2,963	0.0004	0.0121	−0.83	7.81	3,202***
VN30	M30	21,942	0.0001	0.0039	−1.62	36.16	1,014,769***
VN30	H1	13,523	0.0001	0.0051	−1.33	18.20	134,167***
VN30	H4	5,410	0.0001	0.0084	−1.17	11.23	16,490***
VN100	Daily	2,963	0.0004	0.0119	−0.96	8.21	3,804***
VN100	M30	21,942	0.0000	0.0038	−1.83	37.63	1,108,959***
VN100	H1	13,523	0.0001	0.0050	−1.50	19.36	155,896***
VN100	H4	5,410	0.0002	0.0082	−1.30	11.76	18,833***
VNINDEX	Daily	2,963	0.0004	0.0115	−0.98	8.27	3,898***
VNINDEX	M30	21,942	0.0000	0.0037	−2.04	40.64	1,310,762***
VNINDEX	H1	13,523	0.0001	0.0049	−1.56	19.54	159,748***
VNINDEX	H4	5,410	0.0002	0.0080	−1.31	11.99	19,767***
P _{cap}	Daily	2,963	0.0004	0.0124	−1.07	8.15	3,837***
P _{cap}	M30	21,942	0.0000	0.0041	−1.87	33.94	887,934***
P _{cap}	H1	13,523	0.0001	0.0052	−1.51	18.81	145,968***
P _{cap}	H4	5,410	0.0002	0.0086	−1.33	11.69	18,632***

Notes: Kurt. is non-excess kurtosis; P_{cap} is the mid-cap proxy of Section 2.2. *** $p < 0.01$ (Jarque–Bera). Source: Authors’ calculations based on HOSE index data (TradingView).

At the daily frequency, standard deviations are highest for P_{cap} (0.0124) and VN30 (0.0121) and lowest for VNINDEX (0.0115), consistent with the diversification of the broad index. Across intraday intervals, the standard deviation increases with bar length. All series are negatively skewed, and the Jarque–Bera statistics rise from 3,202–3,898 at 1D to 887,934–1,310,762 at M30, driven by extreme kurtosis (above 33 at M30). This leptokurtosis supports the use of DCCA and MF-DCCA, which do not rely on Gaussian assumptions.

3.2 Average Cross-Correlation Across Trading Frequencies

Table 2 reports the average DCCA cross-correlation coefficients by trading frequency for the nested pairs and for the purged pair.

Table 2 Average DCCA cross-correlation by trading frequency

Trading frequency	Nested group	$P_{\text{cap-VN30}}$	Gap	N
Panel A: Synchronized Common Baseline Window (2017–2024)				
1D (Daily)	0.980	0.890	0.090	1,983
M30 (30-Minute)	0.979	0.890	0.089	19,463
H1 (1-Hour)	0.979	0.891	0.089	9,879
H4 (4-Hour)	0.980	0.892	0.089	3,953
Panel B: Full Historical Depth Sample (2014–2025)				
1D (Daily)	0.977	0.884	0.093 [0.065, 0.124]	2,963
M30 (30-Minute)	0.979	0.883	0.096 [0.070, 0.126]	21,942
H1 (1-Hour)	0.975	0.889	0.086 [0.060, 0.116]	13,523
H4 (4-Hour)	0.976	0.890	0.087 [0.061, 0.114]	5,410

Notes: Averages of $\rho_{\text{DCCA}}(s)$ over $s \leq s_{\text{rel}}$ (Table 3), $m = 1$; nested group = mean of the three parent–child pairs. Brackets: block-bootstrap 95% intervals. Source: Authors’ calculations.

Across both samples, the original nested group maintains an average correlation of 0.975–0.980 at every frequency. In contrast, $P_{\text{cap-VN30}}$ has a lower correlation of 0.883–0.892, a gap of 0.086–0.096 correlation points whose bootstrap intervals (0.060 to 0.126 across frequencies) exclude zero. The purged correlation nevertheless remains high, so the overlap inflates rather than creates the co-movement. Averages in the synchronized window (Panel A) and the full sample (Panel B) differ by at most 0.007 for every series.

Figure 2 plots the DCCA coefficient against the timescale at all four trading frequencies for the purged mid-cap proxy and the nested index pairs.

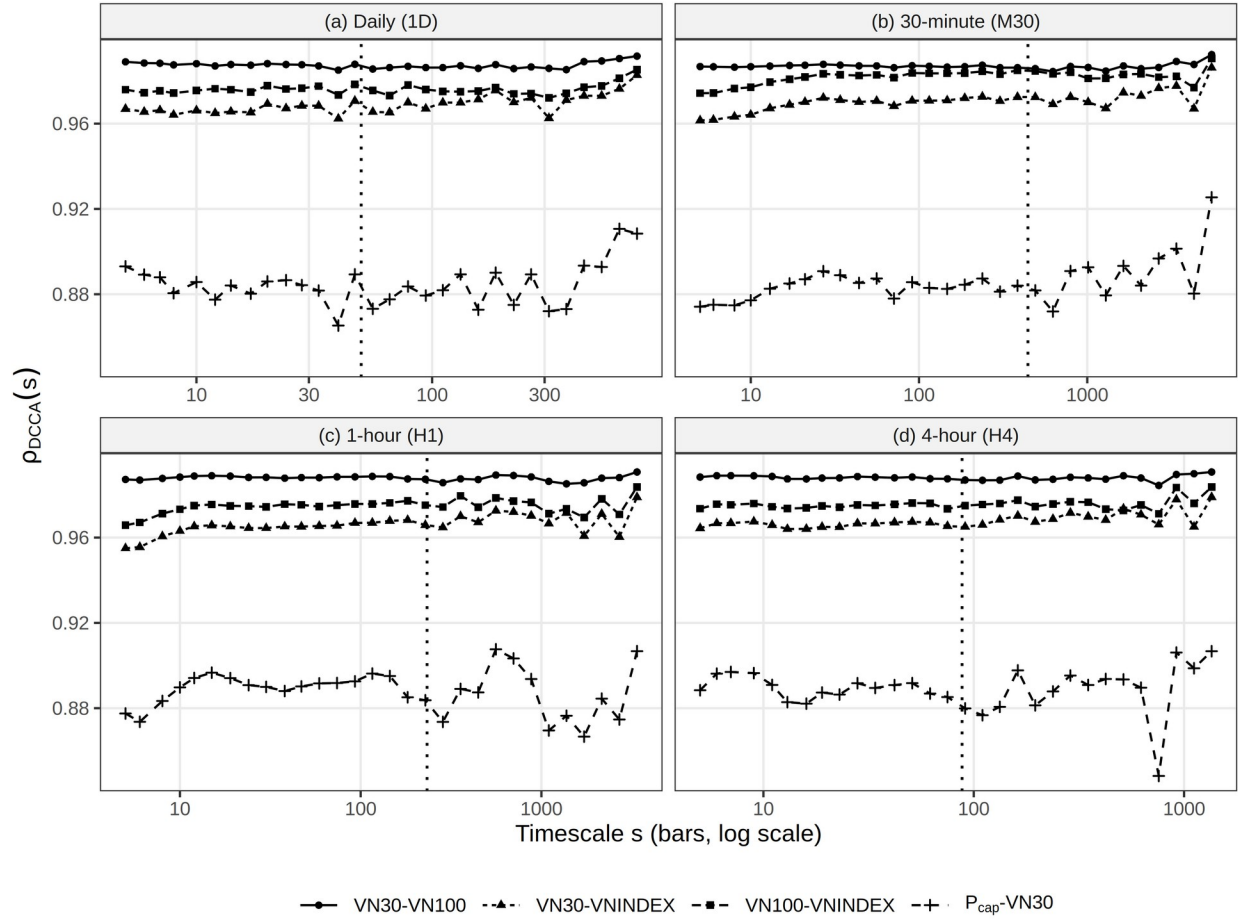


Fig. 2 Multiscale DCCA cross-correlation curves for the nested pairs and $P_{\text{cap}}\text{-VN30}$: (a) 1D, (b) M30, (c) H1, (d) H4

Notes: Line and marker types identify the pairs (legend); vertical dotted lines mark the reliability thresholds s_{rel} (Table 3). Source: Authors' calculations based on HOSE index data (TradingView).

3.3 Finite-Sample Reliability Thresholds

Because the number of segments shrinks as the timescale grows, the DCCA coefficient becomes unreliable at large timescales. For each sample size, we simulate pairs of Gaussian white-noise series with known correlations of -0.3 , 0 , 0.3 , 0.5 , 0.7 and 0.9 (167 replications each, 1,002 per frequency) and compute the mean absolute estimation error on 40 log-spaced scales. The reliability threshold is the largest scale below the first scale at which the worst-case error exceeds 0.05. Because returns are heavy-tailed and volatility-clustered, we repeat the calibration with generalized autoregressive conditional heteroskedasticity, GARCH(1,1), surrogates driven by Student-t innovations with five degrees of freedom. Table 3 reports both thresholds.

Table 3 Finite-sample reliability thresholds by trading frequency

Trading frequency	Sample size (N)	Gaussian s_{rel}	Heavy-tailed s_{rel}
1D (Daily)	2,963	50	20
M30 (30-Minute)	21,942	444	151
H1 (1-Hour)	13,523	233	86
H4 (4-Hour)	5,410	88	28

Notes: Largest scale at which the worst-case mean absolute error stays below 0.05; Gaussian: 1,002 white-noise simulations; heavy-tailed: 300 GARCH(1,1) simulations with Student-t (5) innovations. Source: Authors' calculations.

The heavy-tailed thresholds are smaller, but averaging over them leaves the results intact (nested pairs 0.975–0.979; P_{cap} –VN30 0.882–0.890). The estimates are also insensitive to the detrending order: for VN30–VNINDEX at the daily frequency, the mean reliable-range correlation is 0.9665, 0.9663 and 0.9667 for $m = 1, 2$ and 3. The statistical proxies (P_{heur} , P_{ratio} , P_{res}) produce average DCCA correlations with VN30 between -0.16 and 0.07 , confirming that they are residual artifacts, whereas P_{cap} correlates with them at only 0.277 – 0.486 , indicating that it captures a distinct mid-cap signal.

3.4 Multifractal Structure

The generalized Hurst exponent at the second order, $h(2)$, lies within 0.527 – 0.544 for all series and frequencies, indicating weak positive persistence. The range of generalized exponents is 0.252 – 0.431 for the nested pairs and 0.435 – 0.656 for P_{cap} –VN30, but shuffled surrogates show that much of this range reflects fat tails rather than temporal cross-correlation: the P_{cap} –VN30 range exceeds the 95th surrogate percentile only at 1D (observed 0.656 against a 95th percentile of 0.577), and the nested pairs exceed it in 2 of 12 cases. The multifractal evidence therefore supports only a modest, mainly daily, nonlinear cross-correlation structure in the purged series.

Figure 3 contrasts the multifractal spectra and generalized exponents of the purged mid-cap proxy with those of the nested pairs at the daily frequency.

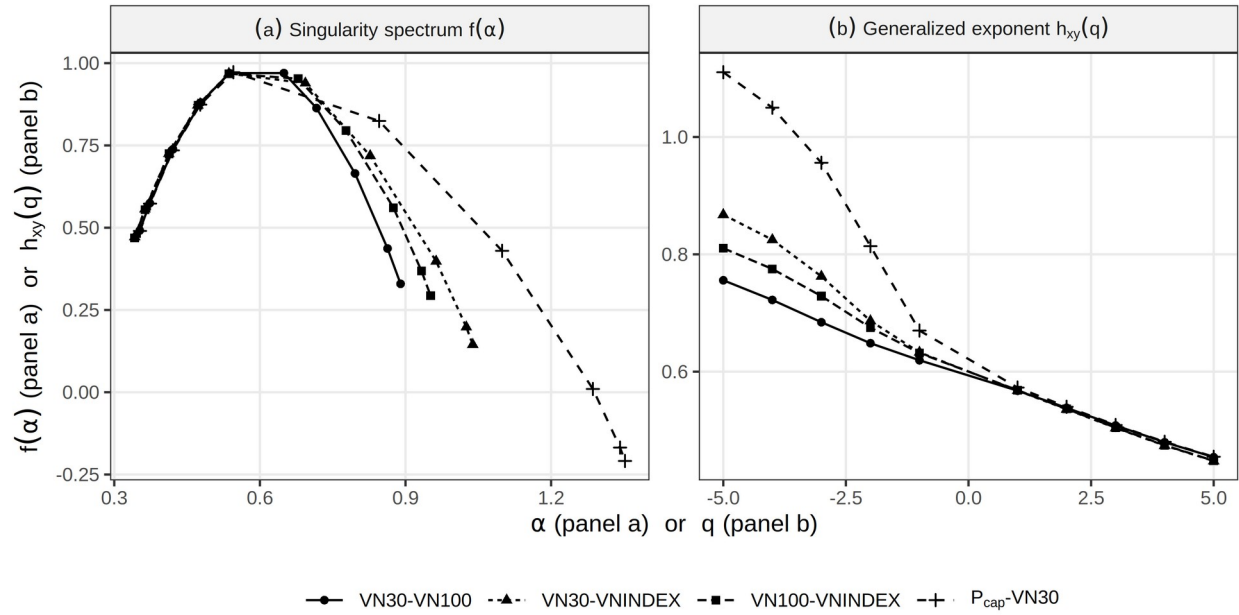


Fig. 3 MF-DCCA at the daily frequency: (a) singularity spectra $f(\alpha)$; (b) generalized exponents $h_{xy}(q)$

Notes: $q \in [-5, 5] \setminus \{0\}$; daily data. Source: Authors' calculations based on HOSE index data (TradingView).

3.5 Volatility Regimes and the Forbes–Rigobon Adjustment

Table 4 reports the unadjusted and Forbes–Rigobon conditioned correlations for the chronological and rolling-volatility regimes.

Table 4 Cross-correlation across volatility regimes and Forbes–Rigobon conditioning

Asset pair	ρ_{low}	ρ_{high}	δ	ρ^*	$\rho^* - \rho_{low}$ [95% CI]	p-value	Power
Panel A: Chronological Crisis							
P_{cap} -VN30	0.847	0.924	2.21	0.803	-0.044 [-0.082, 0.001]	0.985	0.77
Panel B: 20-Day Rolling Regimes							
P_{cap} -VN30	0.727	0.928	7.06	0.661	-0.066 [-0.132, 0.008]	0.974	0.41

Notes: Pearson correlations of daily returns; δ = relative increase in VN30 return variance; one-sided bootstrap p-value for $H_0: \rho^* \leq \rho_{low}$; power to detect an increase of 0.05. Panel A: 1,000 calm and 539 crisis days; Panel B: 739 days per regime. Source: Authors' calculations.

The formal test in Table 4 is restricted to the disjoint pair P_{cap} –VN30, because nested pairs embed the conditioning asset in the dependent series and violate the exogeneity condition of Forbes and Rigobon (2002). For diagnostic comparison only, raw correlations for the nested pairs rise from 0.959–0.983 in calm periods to 0.978–0.992 in crisis episodes, and from 0.921–0.973 to 0.981–0.992 across rolling-volatility regimes.

In Panel A, raw cross-tier correlation rises from 0.847 in calm years to 0.924 during the three crisis episodes, while the variance of VN30 returns increases by 221%. After conditioning, the adjusted crisis correlation is 0.803, and the difference from the calm level (–0.044; 95% interval –0.082 to 0.001) provides no evidence of a structural increase ($p = 0.985$). Because the interval also includes zero, the data do not support a structural decrease either. The power to detect an increase of 0.05 is 0.77, so moderate structural increases are unlikely, whereas small ones cannot be excluded.

Panel B sorts days by the rolling 20-day volatility of VNINDEX, producing a larger variance expansion ($\delta = 7.06$). The raw correlation rises from 0.727 to 0.928, but the adjusted crisis correlation (0.661) again does not exceed the calm level (difference –0.066, 95% interval –0.132 to 0.008; $p = 0.974$), although this test has low power (0.41). Both definitions therefore point to interdependence amplified by volatility rather than contagion, a conclusion that rests on the absence of evidence for an increase rather than on proof of its absence. Because the regimes are sorted on VNINDEX, which contains the mid-caps, and are classified ex post, the comparison describes in-sample regimes rather than a real-time signal.

3.6 Markowitz Portfolio Variance Implications

Under the Markowitz (1952) mean-variance framework, the variance of a two-asset portfolio with arbitrary asset volatilities is:

(16)

When an investor applies a static full-sample correlation instead of a regime-specific correlation, the relative percentage error in portfolio variance for an equally weighted portfolio () is:

(17)

All figures use regime-specific volatilities and Pearson correlations in the exact expression, so the static and regime-specific cases differ only in the correlation. The errors differ between tradable cash portfolios and factor exposures.

For cash portfolios formed from parent and child indices (such as VN30–VN100), co-movement is dominated by constituent overlap. The static daily correlation of 0.988 compares with rolling-regime correlations of 0.973 and 0.992, so the static model overstates portfolio variance by 0.77% in the calm regime and understates it by 0.20% in the turbulent regime. Across the three parent–child pairs and both regime definitions, the largest absolute error is 2.35%, which is negligible for operational risk management.

For the mid-cap factor exposure, the errors are larger. The static model overstates portfolio variance by 2.23% (95% interval 0.91 to 3.94) in the chronological calm regime and by 9.38% (6.20 to 13.46) in the rolling low-volatility regime, whose quietest days exaggerate the effect. Because P_{cap} is a non-investable construct, these figures describe the risk of a mid-cap factor exposure (for example, one obtained through a mid-cap index fund) combined with large caps, not a portfolio that can be held directly.

In the turbulent regime, the static model understates portfolio variance by 1.84% (chronological) and 2.07% (rolling). Although modest, these errors arise when the variance of VN30 returns has

risen by 221% to 706% (Table 4). All regimes are identified ex post, so the figures measure the in-sample cost of ignoring regime dependence rather than the performance of a real-time risk-budgeting rule.

3.7 Multiscale Scaling and Economic Mechanisms

Table 5 reports the scaling slopes of the DCCA coefficients at the M30 frequency over the full and reliable scale ranges, with block-bootstrap intervals.

Table 5 Log-linear scaling regressions of DCCA coefficients (M30)

Pair	Full-range slope	95% CI	Reliable-range slope	95% CI
VN30–VNINDEX	0.0016	[−0.0025, 0.0034]	0.0022	[0.0007, 0.0038]
VN100–VNINDEX	0.0009	[−0.0010, 0.0021]	0.0022	[0.0011, 0.0032]
VN30–VN100	0.0001	[−0.0015, 0.0010]	−0.0001	[−0.0007, 0.0007]
P_{cap} –VN30	0.0025	[−0.0102, 0.0093]	0.0017	[−0.0045, 0.0078]
P_{heur} –VN30	0.0222	[−0.0236, 0.0582]	0.0289	[−0.0033, 0.0534]
P_{ratio} –VN30	0.0221	[−0.0226, 0.0589]	0.0291	[−0.0030, 0.0531]
P_{res} –VN30	0.0220	[−0.0249, 0.0574]	0.0283	[−0.0029, 0.0526]

Notes: Slopes of $\rho_{\text{DCCA}}(s)$ on $\ln s$; full range: 30 scales from 5 to 5,485 bars; reliable range: $s \leq 444$. Brackets: block-bootstrap 95% intervals (499 replications). Source: Authors' calculations.

The resampling results show three patterns.

First, for the nested pair VN30–VN100, the slope is close to zero in both ranges: because VN30 constituents dominate VN100 capitalization, their mechanical co-movement is invariant to the timescale.

Second, the pairs involving the broad market rise significantly with the horizon within the reliable range: VN30–VNINDEX (0.0022, 95% interval [0.0007, 0.0038]) and VN100–VNINDEX (0.0022, [0.0011, 0.0032]). Over the reliable M30 range (5 to 444 bars) this corresponds to a rise of about 0.010 in the correlation. Over the full range, where long scales rest on few segments, the intervals include zero.

Third, the purged mid-cap correlation shows no significant horizon dependence: the P_{cap} –VN30 slope is 0.0025 [−0.0102, 0.0093] over the full range and 0.0017 [−0.0045, 0.0078] over the

reliable range. The statistical residual proxies have larger but imprecise slopes, consistent with amplified noise.

The horizon dependence of the broad-market pairs is consistent with two complementary mechanisms. Under the Epps (1979) effect, non-synchronous trading in less liquid constituents dampens correlations over the shortest intervals, roughly the first ten bars, and temporal aggregation removes this dampening. Beyond that range, gradual information diffusion (Hong and Stein 1999) can sustain a further rise if prices of different constituents adjust to common news at different speeds. That the purged mid-cap series shows no significant rise suggests that these effects are weaker than index-level sampling noise once the overlap is removed.

Index-level closing data cannot distinguish microstructure frictions from gradual information diffusion; separating the two would require tick-by-tick order-book data and firm-level news timestamps.

3.8 Robustness Checks

3.8.1 Scaling Slopes at the Other Frequencies

Table 6 reports the same slopes at the daily, 1-hour and 4-hour frequencies for the broad-market pair VN30–VNINDEX and the purged pair P_{cap} –VN30.

Table 6 Scaling slopes of DCCA coefficients at the daily, 1-hour and 4-hour frequencies

Frequency and pair	Full-range slope [95% CI]	Reliable-range slope [95% CI]	Assessment
Daily: VN30–VNINDEX	0.0020 [−0.0064, 0.0043]	0.0010 [−0.0031, 0.0043]	Flat
Daily: P_{cap} –VN30	0.0014 [−0.0255, 0.0118]	−0.0040 [−0.0157, 0.0085]	Flat
1-hour: VN30–VNINDEX	0.0015 [−0.0027, 0.0040]	0.0024 [0.0006, 0.0041]	Rises (reliable range)
1-hour: P_{cap} –VN30	−0.0003 [−0.0137, 0.0089]	0.0020 [−0.0035, 0.0089]	Flat
4-hour: VN30–VNINDEX	0.0014 [−0.0056, 0.0039]	0.0003 [−0.0026, 0.0029]	Flat
4-hour: P_{cap} –VN30	0.0001 [−0.0176, 0.0103]	−0.0020 [−0.0124, 0.0067]	Flat

Notes: Block-bootstrap 95% intervals (499 replications); reliable ranges from Table 3 (Gaussian).

Source: Authors' calculations.

At the 1-hour frequency, VN30–VNINDEX again rises significantly within the reliable range (0.0024, [0.0006, 0.0041]), whereas at the daily and 4-hour frequencies no slope is distinguishable from zero. The P_{cap} –VN30 slope is insignificant at every frequency. The horizon dependence documented in this study is therefore confined to broad-market pairs at the two shortest intraday frequencies.

3.8.2 Sensitivity of P_{cap} to the Capitalization Weight

Because constituent capitalizations drift with relative price movements and periodic rebalancing, Table 7 re-estimates the key P_{cap} statistics for large-cap weights from 0.60 to 0.75.

Table 7 Sensitivity of P_{cap} to the capitalization weight w_{30}

Weight (w_{30})	Mean ρ_{DCCA} (1D)	Slope β_{scale} (M30, full range)	Calm ρ_{low}	Crisis ρ_{high}
0.6000	0.924	0.0014	0.899	0.952
0.6500	0.903	0.0019	0.871	0.937
0.6826 (baseline)	0.884	0.0025	0.847	0.924
0.7200	0.855	0.0033	0.812	0.904
0.7500	0.824	0.0043	0.775	0.881

Notes: ρ_{low} and ρ_{high} are Pearson correlations in the chronological regimes (Table 4, Panel A); slopes are point estimates (inference in Table 5). Source: Authors' calculations.

Table 7 shows that a higher large-cap weight removes more of the VN30 component and therefore lowers the P_{cap} –VN30 correlation (from 0.924 at a weight of 0.60 to 0.824 at 0.75). The level of the purged correlation is thus conditional on the assumed weight, but the qualitative conclusions are not: in every specification the P_{cap} correlation remains at least 0.05 below the 0.977 average of the nested daily pairs, and the raw crisis tightening (0.053–0.106) remains modest relative to the accompanying volatility expansion.

4 Discussion

4.1 Principal Findings

We examine multiscale cross-correlations and constituent overlap among nested equity benchmarks in Vietnam. Using DCCA and MF-DCCA on intraday and daily data, we separate the effect of shared capitalization from economic co-movement.

Three findings stand out.

First, index pairs that embed large-cap constituents within broader benchmarks exhibit cross-correlations inflated by index engineering. Purging shared constituent capitalization through a market-weighted proxy lowers the correlation by about 0.09, a gap whose bootstrap intervals exclude zero, although the purged co-movement remains high. Horizon dependence is confined to broad-market pairs at the shortest intraday frequencies; the purged mid-cap correlation is statistically flat across horizons.

Second, once volatility is conditioned on, there is no evidence that cross-tier co-movement rises structurally during market turmoil. Systemic stress raises volatility across capitalization tiers, so

diversification benefits shrink when they are most needed even though the underlying dependence appears stable; small structural increases cannot be excluded given the power of the test.

Third, cash portfolios composed of nested parent–child indices experience negligible variance misstatement, whereas static correlations misstate the risk of mid-cap factor exposures: in-sample, they overstate variance by 2.2–9.4% in calm regimes and understate it by about 2% during crises.

4.2 Implications for Capital Markets

These findings carry implications for institutional investors, index providers and regulators in Vietnam; for other ASEAN markets with nested architectures, such as Malaysia, Thailand and Indonesia, they are hypotheses to be tested rather than established results.

For institutional portfolio management, allocators in Vietnam gain little diversification from blending large-cap and broad-market parent–child indices. When constructing multi-capitalization strategies, risk managers should replace static correlation assumptions with volatility-regime-conditioned covariance frameworks that recognize both overstated risk in tranquil periods and elevated co-movement during market distress.

For index design, the HOSE and index providers in Vietnam could promote standalone, non-overlapping segment benchmarks. Exchange-traded funds that track the VNMIDCAP (VN70) index would give institutional investors a mid-cap building block without the hidden overlap of headline index blends.

Finally, futures on mid-cap and sector indices, in addition to the existing VN30 futures, would let institutional investors hedge cross-tier exposure without selling cash positions during market stress.

4.3 Limitations and Future Research

This study has limitations that suggest directions for future research. First, the mid-cap segment is identified through a capitalization-weighted decomposition based on a single factsheet snapshot of free-float weights. Although Section 3.8.2 shows that the conclusions are robust to weights between 0.60 and 0.75, validating the proxy against the exchange-published VNMIDCAP (VN70) index at all four frequencies, and reconstructing the historical VN30 weight from semi-annual constituent reviews, would sharpen identification.

Second, index-level closing prices cannot separate microstructure frictions (Epps 1979) from gradual information diffusion (Hong and Stein 1999); constituent-level tick data would allow these channels to be disentangled. Third, the Forbes and Rigobon (2002) adjustment assumes exogenous large-cap shocks, which forced deleveraging may weaken, and its power against small structural changes is limited; regime-switching or heteroskedasticity-based identification could provide complementary evidence. Fourth, regimes are classified ex post, so an out-of-sample evaluation of regime-conditioned risk budgets is needed before the portfolio results can guide practice. Finally,

applying the framework to the nested index systems of Malaysia, Thailand and Indonesia would test the hypothesis that trading restrictions shape multiscale price synchronization.

4.4 Conclusion

Nested equity indices on the HOSE overstate the co-movement between large caps and the rest of the market because they share constituents. A capitalization-weighted decomposition removes about 0.09 of correlation, the remaining cross-tier dependence shows no structural increase in crises once volatility is accounted for, and static correlations misstate the risk of mid-cap exposures by amounts that depend on the volatility regime. These results favor independent mid-cap benchmarks and risk budgets that condition on the volatility regime.

Ethical standards This study uses only publicly available historical index price data from the Ho Chi Minh City Stock Exchange; it involves no human participants or personal data, therefore required no ethical approval, and complies with the current laws of Vietnam.

Data availability The index price data were obtained from TradingView (exchange: HOSE) and are subject to the vendor's terms of use; the merged dataset is available from the corresponding author upon reasonable request.

Code availability The R code that reproduces every table and figure is provided as Online Resource 1 and will be deposited in a public repository upon acceptance.

Funding Funding information is provided on the separate title page in accordance with the journal's double-blind review policy.

Conflict of interest The authors declare that they have no conflict of interest.

Declaration of generative AI use During the preparation of this work, the authors used an AI assistant (Claude, Anthropic) for language editing, reference verification and translation of the analysis code into R. The authors reviewed all content and take full responsibility for the publication.

References

- Al Rababa'a, A. R., Alomari, M., & McMillan, D. (2021). Multiscale stock-bond correlation: Implications for risk management. *Research in International Business and Finance*, 58, 101435. <https://doi.org/10.1016/j.ribaf.2021.101435>
- Ang, A., & Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443–494. [https://doi.org/10.1016/S0304-405X\(02\)00068-5](https://doi.org/10.1016/S0304-405X(02)00068-5)
- Aslam, F., Ferreira, P., Ali, H., Arifa, & Oliveira, M. (2023). Islamic vs. conventional equity markets: A multifractal cross-correlation analysis with economic policy uncertainty. *Economies*, 11(1), 16. <https://doi.org/10.3390/economies11010016>
- Chen, Y., Zhang, J., Lu, L., & Xie, Z. (2024). Cross-correlation and multifractality analysis of the Chinese and American stock markets based on the MF-DCCA model. *Heliyon*, 10(15), e36537. <https://doi.org/10.1016/j.heliyon.2024.e36537>
- Epps, T. W. (1979). Comovements in stock prices in the very short run. *Journal of the American Statistical Association*, 74(366), 291–298. <https://doi.org/10.1080/01621459.1979.10482508>

- Forbes, K. J., & Rigobon, R. (2002). No contagion, only interdependence: Measuring stock market comovements. *The Journal of Finance*, 57(5), 2223–2261. <https://doi.org/10.1111/0022-1082.00494>
- Hong, H., & Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *The Journal of Finance*, 54(6), 2143–2184. <https://doi.org/10.1111/0022-1082.00184>
- Jiang, Z.-Q., & Zhou, W.-X. (2011). Multifractal detrending moving-average cross-correlation analysis. *Physical Review E*, 84(1), 016106. <https://doi.org/10.1103/PhysRevE.84.016106>
- Kantelhardt, J. W., Zschiegner, S. A., Koscielny-Bunde, E., Havlin, S., Bunde, A., & Stanley, H. E. (2002). Multifractal detrended fluctuation analysis of nonstationary time series. *Physica A: Statistical Mechanics and its Applications*, 316(1–4), 87–114. [https://doi.org/10.1016/S0378-4371\(02\)01383-3](https://doi.org/10.1016/S0378-4371(02)01383-3)
- Karim, B. A., & Ning, H. X. (2013). Driving forces of the ASEAN-5 stock markets integration. *Asia-Pacific Journal of Business Administration*, 5(3), 186–191. <https://doi.org/10.1108/APJBA-07-2012-0053>
- Le, T. T. V., Dang, T. P. T., & Phan, T. H. N. (2025). The nonlinear dependence of the Vietnam stock market on the Asian stock market: Evidence from a quantile-on-quantile regression. *International Journal of Innovative Research and Scientific Studies*, 8(4), 7901–7914. <https://doi.org/10.53894/ijirss.v8i4.7901>
- Lean, H. H., & Teng, K. T. (2013). Integration of world leaders and emerging powers into the Malaysian stock market: A DCC-MGARCH approach. *Economic Modelling*, 32, 333–342. <https://doi.org/10.1016/j.econmod.2013.02.013>
- Longin, F., & Solnik, B. (2001). Extreme correlation of international equity markets. *The Journal of Finance*, 56(2), 649–676. <https://doi.org/10.1111/0022-1082.00340>
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77–91. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>
- Michis, A. A. (2022). Multiscale partial correlation clustering of stock market returns. *Journal of Risk and Financial Management*, 15(1), 24. <https://doi.org/10.3390/jrfm15010024>
- Oświecimka, P., Drożdż, S., Forczek, M., Jadach, S., & Kwapień, J. (2014). Detrended cross-correlation analysis consistently extended to multifractality. *Physical Review E*, 89(2), 022805. <https://doi.org/10.1103/PhysRevE.89.022805>
- Peng, C.-K., Buldyrev, S. V., Havlin, S., Simons, M., Stanley, H. E., & Goldberger, A. L. (1994). Mosaic organization of DNA nucleotides. *Physical Review E*, 49(2), 1685–1689. <https://doi.org/10.1103/PhysRevE.49.1685>
- Podobnik, B., & Stanley, H. E. (2008). Detrended cross-correlation analysis: A new method for analyzing two nonstationary time series. *Physical Review Letters*, 100(8), 084102. <https://doi.org/10.1103/PhysRevLett.100.084102>
- Podobnik, B., Jiang, Z.-Q., Zhou, W.-X., & Stanley, H. E. (2011). Statistical tests for power-law cross-correlated processes. *Physical Review E*, 84(6), 066118. <https://doi.org/10.1103/PhysRevE.84.066118>
- Rodriguez, E., & Alvarez-Ramirez, J. (2021). Time-varying cross-correlation between trading volume and returns in US stock markets. *Physica A: Statistical Mechanics and its Applications*, 581, 126211. <https://doi.org/10.1016/j.physa.2021.126211>
- Tran, M. H., & Tran, N. M. (2025). High-frequency dynamics of the Vietnam stock market. *VNU Journal of Economics and Business*, 5(2), 51–59. <https://doi.org/10.57110/vnu-jeb.v5i2.395>
- Zebende, G. F. (2011). DCCA cross-correlation coefficient: Quantifying level of cross-correlation. *Physica A: Statistical Mechanics and its Applications*, 390(4), 614–618. <https://doi.org/10.1016/j.physa.2010.10.022>
- Zhou, W.-X. (2008). Multifractal detrended cross-correlation analysis for two nonstationary signals. *Physical Review E*, 77(6), 066211. <https://doi.org/10.1103/PhysRevE.77.066211>
- Zhou, W., Huang, J., & Wang, M. (2025). Multifractal characteristics and information flow analysis of stock markets based on multifractal detrended cross-correlation analysis and transfer entropy. *Fractal and Fractional*, 9(1), 14. <https://doi.org/10.3390/fractalfract9010014>